install_certificate

Command: `install_certificate`

Description

This API is used to install certificates.

Download pdf: [Download install_certificate as PDF](https://aa5123.github.io/zebra-rfd40-rfd90-iot-docs/command-pdfs/install_certificate.pdf)

Request

Request Fields

Field	Type	Required	Description	Example
`command`	string	Yes	Command issued to install new certificate	`install_certificate`
`requestId`	string	Yes	A unique identifier for the request, allowing tracking and debugging of the operation.	`abc123`
`certDetails`	object	No		``
`certDetails.name`	string	No	The name of the certificate, typically representing the application or agent.	`sotiAgent`
`certDetails.type`	string (enum)	Yes	The type of the certificate, indicating its purpose or usage (e.g., client, server, etc.).	`client`

Field	Type	Required	Description	Example
`certDetails.authenticationType`	string (enum)	Yes	Specifies the authentication method required. If BASIC, username and password are required. If NONE, no authentication credentials are needed.	`BASIC`
`certDetails.url`	array	No	A list of key-value pairs representing URL parameters.	
`certDetails.authenticateOptions`	object	No	Options for authentication, including username and password.	
`certDetails.authenticateOptions.username`	string	No	The username for authentication when required.	`rfid`
`certDetails.authenticateOptions.password`	string	No	The password for authentication when required.	`rfid`
`certDetails.pfxPassword`	string	No	The password for the PFX (Personal Information Exchange) file used in secure communications.	`pass@123`
`certDetails.headerParams`	array	No	A list of key-value pairs representing HTTP header parameters.	
`certDetails.queryParams`	array	No	A list of key-value pairs representing query parameters.	
`certDetails.verificationType`	string (enum)	No	Specifies the type of verification to perform, such as peer verification or host verification.	`NONE`

Field	Type	Required	Description	Example
`certDetails.caCertificateFileContent`	string	No	The content of the CA (Certificate Authority) certificate file, used for verifying identities.	

Request Examples

WiFi certificate install

```
{
  "command": "install_certificate",
  "requestId": "abc123",
  "certDetails": {
    "name": "sotiAgent",
    "type": "wifi",
    "authenticationType": "CERTIFICATE",
    "url": [
      {
        "key": "ca_cert",
        "value": "http://localhost/ca_cert.pem"
      },
      {
        "key": "client_key",
        "value": "http://localhost/client_key.pem"
      },
      {
        "key": "client_cert",
        "value": "http://localhost/client_cert.pem"
      },
      {
        "key": "cert_key_password",
        "value": "amnin@123"
      }
    ],
    "pfxPassword": "pass@123",
    "verificationType": "VERIFY_HOST",
    "caCertificateFileContent": "----BEGIN---- cert content ----END---"
  }
}
```

Filestore certificate

```
{
  "command": "install_certificate",
```

```
  "requestId": "18996",
  "certDetails": {
    "type": "filestore",
    "authenticationType": "CERTIFICATE",
    "url": [
      {
        "key": "ca_cert",
        "value": "http://scp000555.soticonnect.cloud:4650/cm/api/root-download"
      }
    ],
    "verificationType": "VERIFY_HOST"
  }
}
```

MQTT client certificate

```
{
  "command": "install_certificate",
  "requestId": "19001",
  "certDetails": {
    "name": "mqtt_client",
    "type": "mqtt",
    "authenticationType": "CERTIFICATE",
    "url": [
      {
        "key": "client_cert",
        "value": "https://certserver.company.com/api/client-cert"
      },
      {
        "key": "client_key",
        "value": "https://certserver.company.com/api/client-key"
      },
      {
        "key": "ca_cert",
        "value": "https://certserver.company.com/api/ca-cert"
      }
    ],
    "verificationType": "VERIFY_HOST_PEER"
  }
}
```

Server certificate

```
{
  "command": "install_certificate",
  "requestId": "19002",
```

```
  "certDetails": {
    "name": "secure_server",
    "type": "server",
    "authenticationType": "BASIC",
    "url": [
      {
        "key": "client_cert",
        "value": "https://pki.enterprise.com/download/server.crt"
      },
      {
        "key": "client_key",
        "value": "https://pki.enterprise.com/download/server.key"
      },
      {
        "key": "ca_cert",
        "value": "https://pki.enterprise.com/download/root-ca.crt"
      },
      {
        "key": "cert_key_password",
        "value": "encryptedKeyPassword123"
      }
    ],
    "authenticateOptions": {
      "username": "certadmin",
      "password": "securePass456"
    },
    "verificationType": "VERIFY_PEER"
  }
}
```

Response

Response Fields

Field	Type	Required	Description	Example
`command`	string	Yes		`install_certificate`
`requestId`	string	Yes		`abc123`
`apiVersion`	string (enum)	Yes		

Field	Type	Required	Description	Example
`response`	object	Yes	Status of the command execution — check `code` to confirm success (0) or identify the error.	

Field	Type	Required	Description	Example
`response.code`	integer	Yes	Response code indicating success or failure. • **0** — Success • **1** — Command payload accepted • **2** — Invalid payload • **3** — Not able to retrieve information • **4** — Firmware update in progress • **5** — Can't reboot, inventory in progress • **6** — Region is not configured • **7** — Interface is not available • **8** — Insufficient flash size • **9** — File not found • **10** — Configuration already exists • **11** — Inventory in progress • **12** — No radio operation in progress • **13** — Firmware update failed • **14** — Battery too low for firmware update • **15** — WIFI: SSID not found • **16** — WIFI: Cannot delete active SSID • **17** — WIFI: SSID missing	`0`

Field	Type	Required	Description	Example
			• **18** — WIFI: SSID already exists • **19** — WIFI: SSID count overflow • **20** — WiFi is not supported • **21** — Certificate not found • **22** — Advanced configuration not set • **23** — Invalid enum value • **24** — Max 32 prefilters limit exceeded • **25** — Max 3 publish topics exceeded • **26** — Max 1 subscribe topic exceeded • **27** — Invalid tenant ID length • **28** — Tag match pattern length exceeded	
`response.description`	string	Yes	Human-readable description of the response code (e.g. "Success", "Invalid payload").	`Success`

Response Examples

example1

```
{
  "command": "install_certificate",
  "requestId": "18996",
  "apiVersion": "V1.1",
  "response": {
    "code": 1,
    "description": "Command payload is accepted"
  }
}
```